

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	VIVETHA B	Reg. No.:	192424170
Section / Batch:	B.TECH (AI & DS)	Date:	

Code of Conduct

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VIVETHA B

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

ANSWER

Reference Resolution Algorithm

Reference Resolution is the process of identifying a pronoun or referring expression and linking it to the correct entity mentioned earlier in the discourse.

Algorithm

1. **Input the discourse** and divide it into sentences.
2. **Identify entities** mentioned in the discourse, such as people, objects, and places.
3. **Identify referring expressions**, especially pronouns such as *he, she, it, they*, etc.
4. **Generate candidate antecedents** for each pronoun from previously mentioned entities.
5. **Apply agreement constraints** such as gender and number.
6. **Apply recency**, giving preference to recently mentioned compatible entities.
7. **Apply semantic and discourse context** to determine which candidate makes the most meaningful interpretation.
8. **Select the highest-scoring antecedent** for each pronoun.
9. **Replace the pronouns** with their selected antecedents.
10. **Validate the result** by checking grammatical agreement, semantic meaning, and overall discourse coherence.

Applying the Algorithm

Given:

“Ravi met Arun at the library. He borrowed a book and later returned it.”

The entities introduced are:

- **Ravi** — person
- **Arun** — person
- **library** — place
- **book** — object

The referring expressions are:

- **He**
- **it**

Resolving “He”

Possible antecedents are **Ravi** and **Arun**.

Both Ravi and Arun are masculine and singular, so **gender and number agreement alone cannot decide** the antecedent.

The discourse therefore requires contextual interpretation. Since **Ravi** is the grammatical subject and the main discourse focus of the first sentence, **He** → **Ravi** is the preferred interpretation.

Resolving “it”

The pronoun **it** refers to the most recent compatible object, **book**.

Therefore:

it → **book**

Resolved Discourse

“Ravi met Arun at the library. Ravi borrowed a book and later returned the book.”

Validation of the Resolution

The resolution is coherent because **Ravi** remains the main person performing the actions: he **borrowed** and **returned** the book. The pronoun **it** clearly refers to the **book**, because a book is an appropriate object to borrow and return.

The algorithm uses several discourse cues:

- **Entity identification:** Ravi, Arun, library, and book are recognized.
- **Gender/number agreement:** “He” must refer to a singular masculine person.
- **Discourse focus:** Ravi is the subject and primary focus of the first sentence.
- **Semantic compatibility:** A person can borrow and return a book.
- **Recency:** “it” follows the mention of “a book,” making **book** the strongest antecedent.
- **Coherence:** The final interpretation creates a logical sequence: **Ravi met Arun** → **Ravi borrowed a book** → **Ravi returned the book**.

Thus, the final coreference chains are:

Ravi → **He** → **Ravi**

book → **it** → **book**

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

ANSWER

Algorithm for Text Coherence and Discourse Structure

Text coherence refers to the logical connection between sentences so that the text conveys a meaningful and connected idea. A discourse analysis algorithm identifies entities, keywords, connectives, and semantic relationships to determine relations such as **cause-effect, sequence, and elaboration**.

Algorithm

1. **Input the discourse** and divide it into individual sentences.
2. **Identify important entities and events** in each sentence.
3. **Detect discourse connectives** such as *therefore, because, then, after, and however*.
4. **Analyze temporal relationships** to identify whether one event occurs before or after another.
5. **Analyze causal relationships** to determine whether one event causes another.
6. **Compare consecutive sentences** to identify semantic relationships.
7. **Identify the main topic or situation** connecting the sentences.
8. **Assign discourse relations** such as Cause-Effect, Sequence, or Elaboration.
9. **Construct a discourse tree** showing how the sentences are connected.
10. **Validate coherence** by checking whether the relations form a logical progression.

Application to the Given Discourse

Sentence 1

"The roads were flooded after heavy rainfall."

This introduces the **heavy rainfall → flooded roads** relationship.

Sentence 2

"Therefore, schools were closed for the day."

The word **"Therefore"** explicitly indicates a **Cause-Effect** relationship.

- Cause: Roads were flooded.

- Effect: Schools were closed.

So:

S1 → S2 = Cause-Effect

Sentence 3

“Students attended classes online.”

This describes what happened after schools were closed. Therefore, it has a **Sequence/Result** relationship with Sentence 2.

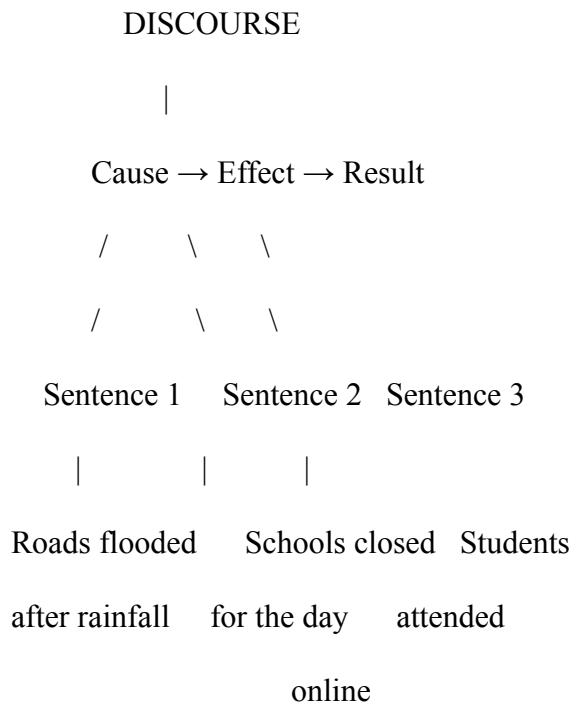
- Schools were closed.
- As a consequence, students attended classes online.

So:

S2 → S3 = Sequence/Result

Discourse Structure Representation

A simple discourse tree can be represented as:



Another compact representation is:

S1: Heavy rainfall

↓

S1: Roads were flooded

↓ Cause

S2: Schools were closed

↓ Result / Sequence

S3: Students attended classes online

Discourse Relations

1. **Heavy rainfall** → **flooded roads**: Cause-Effect
2. **Flooded roads** → **schools closed**: Cause-Effect
3. **Schools closed** → **online classes**: Result/Sequence

Overall, the discourse follows a clear **cause** → **consequence** → **response** structure.

How the Relations Maintain Coherence

The discourse is coherent because each sentence is logically connected to the previous one.

- The **first sentence** establishes the initial situation: heavy rainfall causes flooding.
- The **second sentence** explains the consequence of the flooding. The connective “**Therefore**” makes this causal relationship explicit.
- The **third sentence** describes the resulting action taken by students after schools are closed.
- The sequence follows a natural real-world progression:

Heavy rainfall → **Roads flooded** → **Schools closed** → **Online classes**

Thus, no sentence appears unrelated or disconnected. The **cause-effect and sequence relations** create a continuous chain of events, allowing the reader to easily understand why each event happens and how one situation leads to the next.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

ANSWER

Algorithm for Dialogue Act Identification

A **Dialogue Act** represents the communicative function of an utterance, such as making a request, asking a question, providing information, confirming something, or performing an action.

Algorithm

1. **Input the conversation** as a sequence of user and agent utterances.
2. **Separate each utterance** based on the speaker.
3. **Identify keywords and sentence patterns**, such as *Can you*, *where*, *I want*, *sure*, and *booked*.
4. **Determine the communicative intention** of each utterance.
5. **Classify the utterance** into Request, Question, Inform, Confirmation, or Action.
6. **Use conversation context** to resolve ambiguous utterances.
7. **Create the dialogue-act sequence** in chronological order.
8. **Check whether each act logically connects** with the previous act.

Applying the Algorithm

1. User: “Can you book a train ticket for me?”

Dialogue Act: Request

The user is asking the agent to perform a task—booking a train ticket.

Intent: Book a train ticket.

2. Agent: “Sure, where would you like to travel?”

Dialogue Act: Question

The agent asks for missing information required to complete the booking.

Intent: Obtain the destination.

3. User: “I want to go to Chennai.”

Dialogue Act: Inform

The user provides the requested information.

Intent: Inform the agent that the destination is Chennai.

4. Agent: “Your ticket has been booked.”

Dialogue Act: Confirmation

The agent confirms that the requested booking has been completed.

Intent: Confirm successful completion of the requested action.

Dialogue-Act Sequence

User: Request



Agent: Question



User: Inform



Agent: Confirmation

Therefore:

Request → Question → Inform → Confirmation

Evaluation

The identified dialogue acts help the conversational agent understand the user's intention and manage the conversation systematically.

- **Request** tells the agent that the user wants an action to be performed.
- **Question** allows the agent to collect missing information needed for the action.
- **Inform** provides the requested information, in this case, the destination **Chennai**.
- **Confirmation** indicates that the requested task has been completed.

The sequence also maintains **dialogue coherence** because every agent response is related to the user's previous utterance. The agent first recognizes the booking request, asks for the required destination, receives the information, and then confirms completion.

Thus, dialogue-act classification allows a conversational agent to convert natural-language utterances into structured intentions and respond appropriately at each stage of the interaction.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

ANSWER

Algorithm for Language Generation and Surface Realization

Language Generation converts a **semantic representation** into a grammatically correct natural-language sentence. The process involves **lexical selection, sentence structuring, grammatical realization, and final validation**.

Algorithm

1. **Input the semantic representation** containing the action, agent, object, and tense.
2. **Perform lexical selection** by mapping semantic concepts to suitable words.
3. **Determine the sentence structure** using the selected agent, action, and object.
4. **Apply tense information** to change the verb into its required grammatical form.
5. **Apply agreement rules** between the subject and verb.
6. **Arrange the words** according to the target language's grammatical order.
7. **Perform surface realization** by adding required articles, punctuation, and morphological changes.
8. **Validate the generated sentence** for grammar, tense, word order, and semantic consistency.
9. **Output the final natural-language sentence.**

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

1. Lexical Selection

The semantic concepts are mapped to appropriate English words:

- **Agent** → **Student**
- **Action** → **Buy**
- **Object** → **Book**
- **Tense** → **Past**

The past-tense form of **buy** is **bought**.

Therefore:

Student + bought + book

An appropriate article is added before the singular countable nouns:

The student + bought + a book

2. Sentence Structuring

English generally follows the **Subject–Verb–Object (SVO)** structure.

Subject	Verb	Object
---------	------	--------

↓

↓

↓

The student	bought	a book
-------------	--------	--------

So the basic structure becomes:

The student bought a book.

3. Surface Realization

During surface realization, the semantic structure is converted into the final grammatical form.

- **Buy** → **bought** because the tense is past.
- **Student** → **The student** to form a natural noun phrase.
- **Book** → **a book** because *book* is a singular countable noun being introduced.
- Words are arranged in **SVO order**.
- A period is added at the end.

Final Sentence

The student bought a book.

4. Grammatical Validation

The generated sentence is grammatically correct because:

1. **Subject:** “The student” is a valid noun phrase.
2. **Verb:** “Bought” is the correct past-tense form of “buy.”
3. **Object:** “A book” is a grammatically correct noun phrase.
4. **Word order:** The sentence follows English **Subject–Verb–Object** order.
5. **Tense:** The sentence correctly expresses a **past action**.
6. **Agreement:** “The student” is singular and the sentence uses an appropriate verb form.

7. **Meaning:** The generated sentence preserves the original semantic representation: **Student + Buy + Book + Past.**

Overall Generation Flow

Semantic Representation



Lexical Selection



Student + Buy + Book



Tense Realization



The student + bought + a book



Sentence Structuring



The student bought a book.



Grammatical Validation



Final Output

Thus, the language-generation algorithm successfully transforms the semantic representation into the grammatically correct sentence **“The student bought a book.”**

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.

2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

ANSWER

Algorithm: Hybrid Interlingua + Statistical Machine Translation

A **hybrid machine translation system** combines the linguistic understanding of an **Interlingua-based approach** with the data-driven selection ability of **Statistical Machine Translation (SMT)**.

Algorithm

1. **Input the source sentence.**
2. **Perform lexical and syntactic analysis** to identify the subject, verb, object, tense, and other grammatical information.
3. **Create an Interlingua representation** that captures the language-independent meaning.
4. **Generate candidate translations** in the target language from the Interlingua.
5. **Calculate statistical scores** for each candidate using translation probability and language-model probability.
6. **Select the highest-scoring candidate** that also satisfies the Interlingua meaning.
7. **Perform surface realization** to produce a grammatically correct target sentence.
8. **Validate the output** for meaning preservation, grammar, and fluency.

1. Analyze the Source Sentence

Source sentence:

“The boy is playing football.”

The linguistic analysis is:

- **Subject:** the boy
- **Verb:** is playing
- **Object:** football
- **Tense:** Present continuous
- **Action:** Playing
- **Agent:** Boy
- **Object/Activity:** Football
- **Number:** Singular

The basic semantic structure is:

Agent = Boy

Action = Play

Object = Football

Tense = Present Continuous

2. Create an Interlingua Representation

The Interlingua represents the meaning independently of a particular language.

PLAY(

AGENT = BOY,

OBJECT = FOOTBALL,

TENSE = PRESENT_CONTINUOUS

)

A more detailed representation is:

EVENT: Playing

|

└─ Agent: Boy

└─ Activity/Object: Football

└─ Tense: Present Continuous

This representation separates **meaning** from the English words used to express that meaning.

3. Generate Candidate Target-Language Translations

Assume the target language is **Tamil**.

The Interlingua can generate candidates such as:

1. சிறுவன் கால்பந்து விளையாடுகிறான்.
2. பையன் கால்பந்து விளையாடுகிறான்.
3. சிறுவன் கால்பந்து விளையாடினான்.

Their meanings are approximately:

- Candidate 1: *The boy is playing football.*
- Candidate 2: *The boy is playing football.*

- Candidate 3: *The boy played football.*

Candidate 3 has the wrong tense because the source uses **present continuous**, while Candidate 1 and Candidate 2 preserve the intended meaning.

4. Apply Statistical Scoring

The system can calculate a combined score:

Final Score =

$\alpha \times$ Interlingua Meaning Score

+

$\beta \times$ Translation Probability

+

$\gamma \times$ Language Model Score

For example:

Candidate	Meaning Match	Statistical/Fluency Score	Final Decision
சிறுவன் கால்பந்து விளையாடுகிறான்.	High	High	✓
பையன் கால்பந்து விளையாடுகிறான்.	High	High	Possible
சிறுவன் கால்பந்து விளையாடினான்.	Low	High	✗

The first candidate is selected because it preserves the **agent, action, object, and present-tense meaning** while also being a natural Tamil sentence.

5. Final Translated Sentence

Tamil:

சிறுவன் கால்பந்து விளையாடுகிறான்.

English meaning:
The boy is playing football.

Overall Translation Flow

English Source

"The boy is playing football."



Linguistic Analysis



Interlingua Representation



PLAY(BOY, FOOTBALL, PRESENT_CONTINUOUS)



Generate Tamil Candidates



Statistical Scoring



Select Best Candidate



சிறுவன் கால்பந்து விளையாடுகிறான்.

Evaluation

Advantages of Interlingua

1. **Preserves meaning:** The Interlingua explicitly records the boy as the agent, playing as the action, football as the object, and present continuous as the tense.
2. **Reduces ambiguity:** The system does not depend only on individual word-to-word translations.
3. **Language independence:** The same semantic representation can be used to generate translations into multiple languages.
4. **Preserves grammatical information:** Important features such as tense, number, and semantic roles can be maintained.

Advantages of Statistical Translation

1. **Improves fluency:** Statistical models can select target-language expressions that are commonly used.
2. **Handles lexical alternatives:** For example, both சிறுவன் and பையன் may represent “boy,” and statistical evidence can help select the more appropriate expression.
3. **Uses target-language patterns:** The model can prefer naturally occurring Tamil word combinations.
4. **Ranks candidates:** Instead of accepting the first possible translation, it selects the candidate with the strongest combined score.

Overall Benefit

The **Interlingua component ensures semantic accuracy**, while the **statistical component improves fluency and naturalness**. Combining both approaches reduces ambiguity and produces translations that are both **meaning-preserving and natural-sounding**.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____